S.A. ENGINEERING COLLEGE, CHENNAI-77

(Autonomous-Institution Level Research Centre, Affiliated to Anna university)

DEPARTMENT OF MATHEMATICS

PROBLEMS FOR PRACTISE -WEEKLY ASSIGNMENT -I

Sub.code/Sub.Name:MA3403-OPTIMIZATION TECHNIQUES
Year/Dept./Sem:II/IT/IV **Date:2.1.2026**

UNIT -I-OPTIMIZATION TECHNIQUES

1.A firm produces 3 products. These products are processed on 3 different machines. The time required to manufacture one unit of each of the 3 products and the daily capacity of the 3 machines are given below
It is required to determine the number of units to be manufactured for each product daily.The profit per unit for product 1,2 and 3 is Rs4, Rs3, Rs6 respectively. It is assumed that all the amounts produced are consumed in the market. Formulate the mathematical model for the problemand solve the LPP by graphical method.

Machine	Time per unit (minutes)			Machine Capacity (Minutes /day)
	Product1	Product2	Product3	
M_1	2	3	2	440
M_2	4	-	3	470
M_3	2	5	-	430

2. Garden Ltd. Has two product Rose and Lotus. To produce one unit of Rose, 2 units of material X and 4 units of material Y are required. To produce one unit of Lotus, 3 units of material X and 2 units of material Y are required. At least 16 units of each material must be used in order to meet the committed sales of Rose and Lotus Cost per unit of material X and material Y are Rs. 2.50 per unit and Rs.0.25 per unit respectively.

You are required:

(i)To formulate mathematical model

(ii)To solve it for the minimum cost(Graphically)

3.Solve by Graphical method

Maximize $z = 100x_1 + 80x_2$ Subject to

$5x_1 + 20x_2 \leq 50,$
$8x_1 + 2x_2 \geq 16,$
$3x_1 - 2x_2 \geq 6,$
$x_1, x_2 \geq 0$

4.Solve by Simplex Method:

Maximize $Z = 4x_1 + 10x_2$

sub to $2x_1 + x_2 \leq 50$

$2x_1 + 5x_2 \leq 100$

$2x_1 + 3x_2 \leq 90$

$x_1, x_2 \geq 0$

5.Solve by Simplex Method:

Maximize $Z = 3x_1 + 6x_2 + 3x_3$

Sub to $3x_1 + 4x_2 + x_3 \leq 3$

$x_1 + 3x_2 + x_3 \leq 1$

$x_1, x_2, x_3 \geq 0.$

Staff In-Charge | R. [signature] 2/1/26 Course Coordinator | R. [signature] 2/1/26 HOD